

LeafLink: Designing Plant-Mediated Multisensory Interaction for Non-Verbal Communication

Shu Wang

University of the Arts London
London college of communication
London, United Kingdom
s.wang0120242@arts.ac.uk

Abstract

LeafLink is a plant-based multisensory interaction system designed to support non-verbal emotional expression for individuals with autism. Grounded in more-than-human HCI, alongside sensory design and art therapy, and informed by recent research that explores non-human and multispecies perspectives, this work investigates how plants can act as soft, living mediators that enable calm and inclusive communication. The system utilizes plant-mediated capacitive sensing to capture human-plant touch and proximity, mapping these interaction data to real-time visual, auditory, and textual feedback. This extended abstract presents the design rationale, system workflow, and insights from field research with an autism charity, co-creation workshops, and prototype evaluations with autistic and neurotypical participants. Preliminary findings indicate that plant-based interactive systems can enhance sensory engagement, promote non-verbal expression, and foster shared attention. Opportunities for future work include exploring lightweight adaptive mapping approaches to better interpret plant-mediated interaction patterns, expanding user studies, and developing adaptive multi-species communication environments.

CCS Concepts

• **Human-centered computing** → Interaction design; Accessibility; Human computer interaction (HCI); HCI theory, concepts and models.

Keywords

more-than-human design, autism, non-verbal communication, plant-based sensory interaction

ACM Reference Format:

Shu Wang. 2026. LeafLink: Designing Plant-Mediated Multisensory Interaction for Non-Verbal Communication. In *Extended Abstracts of the 2026 CHI Conference on Human Factors in Computing Systems (CHI EA '26)*, April 13–17, 2026, Barcelona, Spain. ACM, New York, NY, USA, 5 pages. <https://doi.org/10.1145/3772363.3798314>

1 Introduction

1.1 Problem framing and motivation

Individuals with autism often experience challenges in emotional expression and social communication, particularly in contexts that

prioritize verbal language and efficiency [25]. Many individuals with autism rely on non-verbal cues such as gesture, rhythm, touch, and sensory engagement to communicate feelings and build social connections [2]. However, within mainstream human-computer interaction, communication is still largely framed through language-based interaction, task completion, and rational control. Although inclusive design has gained increasing attention in recent years, non-verbal communication is frequently reduced to functional assistance rather than recognized as a rich and meaningful form of emotional expression [11]. This gap limits how interactive systems support individuals with autism whose communication is embodied, sensory, and relational rather than linguistic.

In this paper, “emotional expression” refers to the embodied articulation of affective states without reliance on verbal language, while “communication” is understood as a relational process that emerges through shared attention and multisensory interaction. LeafLink primarily focuses on supporting emotional expression, with communication arising indirectly through shared plant-mediated interaction.

1.2 Research context and gap

In response to these limitations, HCI research has begun to explore multispecies and more-than-human perspectives that challenge anthropocentric assumptions and expand the notion of interaction beyond human users [9, 17]. Within this context, plants have emerged as a promising interactive medium due to their non-verbal, non-invasive, and sensorial qualities [10, 18]. Prior work has investigated plant-based interaction through visualization, metaphorical feedback, emotional display, and sensing technologies [1, 5, 15, 20]. Nevertheless, much of this research remains symbolic, aesthetic, or one-directional, and rarely addresses non-verbal emotional communication for specific user groups such as individuals with autism. Moreover, plants are often treated as expressive interfaces rather than as active participants within a shared communicative environment. In this work, plant-based systems refer to interactive systems incorporating living plants as sensing and mediating elements, rather than plant-mimicking artifacts.

1.3 Design approach and contribution

This paper introduces LeafLink, a plant-based multisensory interaction system that explores how plants can function as soft mediators of non-verbal emotional expression for individuals with autism. Rather than positioning plants as decorative or metaphorical elements, LeafLink approaches them as responsive co-creators within an interactive ecosystem.



This work is licensed under a Creative Commons Attribution 4.0 International License. *CHI EA '26, Barcelona, Spain*

© 2026 Copyright held by the owner/author(s).

ACM ISBN 979-8-4007-2281-3/2026/04

<https://doi.org/10.1145/3772363.3798314>

Grounded in sensory design, art therapy practices, and more-than-human HCI, the system enables emotional communication through touch, proximity, and shared multisensory feedback. This extended abstract presents the design rationale of LeafLink, its interactive system workflow, and insights from field research with an autism charity, co-creation workshops, and early prototype evaluations. By focusing on calmness, openness, and embodied interaction, this work contributes an early design-led exploration of plant-centered interaction for inclusive, non-verbal communication.

2 Related work

2.1 Multispecies interaction in HCI

Multispecies and more-than-human perspectives in HCI have proposed expanding interaction beyond human users to include animals, plants, and other non-human entities as participants in interactive systems [9, 10, 24]. This body of work has explored interaction as relational, situated, and distributed across human and non-human actors. Design research in this area has foregrounded care, ethics, and ecological relations, offering alternative models of interaction that challenge efficiency-driven paradigms [16]. However, these perspectives have primarily been discussed at a conceptual level, with limited attention to how multispecies interaction might support specific communicative needs or accessibility contexts. Applying a more-than-human perspective in this context allows interaction to move beyond human-centered performance and verbal competence. For individuals with autism, whose communication may be sensory and relational rather than language-dominant, multispecies interaction offers a less pressured and more distributed communicative environment.

2.2 Plant-based interaction in HCI

Plant-based interaction has emerged as a distinct area within multispecies HCI, exploring plants as sensing interfaces, expressive media, and environmental indicators [10, 18]. Prior systems have utilized plant signals for visualization, metaphorical representation, and affective display [1, 3–5, 12, 13, 15, 20, 22, 23]. These works demonstrate the technical and aesthetic potential of plant-centered interaction. Nevertheless, plant-based systems are most often designed as symbolic or observational interfaces, rather than as components of shared, non-verbal communicative environments.

2.3 Non-verbal therapeutic design for children with autism

Art therapy is widely used as a non-verbal intervention method in therapeutic practices and has been shown to support emotional expression and social interaction without reliance on verbal language [6]. Related research on autistic children highlights the value of social games, non-verbal communication training, sensory integration therapy, and interactive tools in supporting social skill development and emotional regulation. In particular, social games facilitate relationship building, and combining verbal and non-verbal communication has been found to further enhance socialization abilities [8].

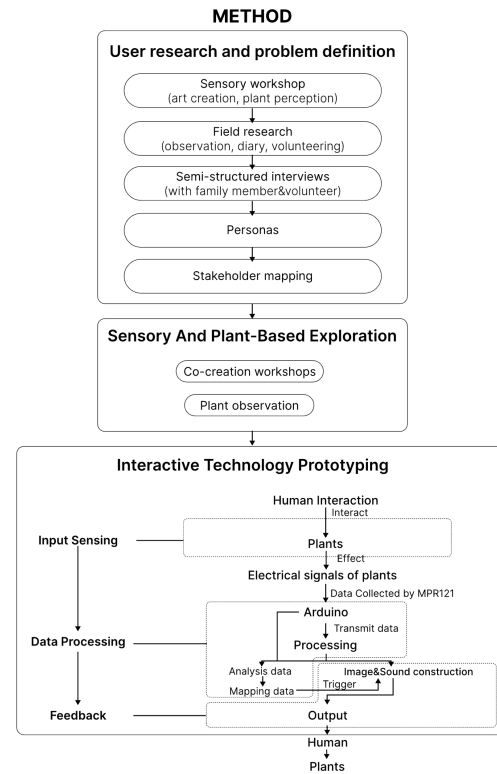


Figure 1: Overall design method (Illustration by the author)

Multisensory approaches have also been explored in therapeutic and supportive contexts. Interaction with plants has been reported to positively influence children’s emotional states and cognitive engagement [21], while sensory garden designs show potential benefits for mental well-being among children with autism [19]. Multisensory stimulation has similarly been examined in dance therapy and co-creative activities, where it has been associated with improved sensory integration, emotional regulation, and increased participation [7, 14].

This body of work suggests opportunities for design-led exploration of non-verbal therapeutic interaction that moves beyond assistive functionality. Integrating non-human elements as part of the therapeutic setting may offer alternative ways to support calm, inclusive, and less directive forms of emotional communication.

3 Design-led Method

LeafLink was developed through a design-led, mixed-methods approach that prioritized exploratory design and experiential insight rather than hypothesis-driven evaluation. The process combined fieldwork, sensory exploration, plant observation, co-creation workshops, and interactive prototyping to iteratively shape the system concept and interaction design. (see Figure 1)

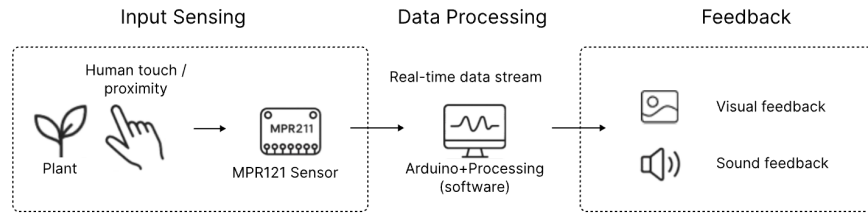


Figure 2: System workflow of plant-mediated multisensory interaction (Illustration by the author)

3.1 Node 1: User Research and Problem Definition

User research was conducted through field engagement with an autism charity, including observation of everyday activities and informal interaction contexts. Semi-structured interviews with a family member of an individual with autism and a volunteer complemented these observations. Personas and a stakeholder map were used to synthesize insights and frame key design considerations, including sensory comfort, emotional safety, and non-verbal reciprocity.

3.2 Node 2: Sensory and Plant-based Exploration

Sensory exploration and co-creation activities were carried out with autistic and neurotypical participants to investigate tactile, visual, and affective responses to plants. Activities such as silent observation and material-based expression informed the system’s visual language and interaction qualities. In parallel, plant observation focused on changes in posture and environmental response, informing subsequent interaction mappings.

3.3 Node 3: Interactive Prototyping

An interactive prototype was developed to explore plant-mediated multisensory interaction. Capacitive sensing was used to capture plant signals, which were mapped to real-time visual, auditory, and textual feedback (see Figure 2). The prototype supported multi-plant interaction, enabling layered and emergent responses shaped by simultaneous human-plant engagement.

4 Design outcomes

4.1 Design Insights from User Research

Early sensory workshops explored participants’ experiences of plants through silent walking and multisensory engagement, including seeing, hearing, and touching. Participants translated these experiences into sketches and wire sculptures, revealing how art-based sensory exploration could serve as a bridge between human perception and plant presence. These activities informed the initial design direction, suggesting that non-verbal, art-informed interaction could support communication between humans and plants.

Field research conducted with Kith & Kids enabled close engagement with participants with autism and volunteers in everyday contexts. Through video documentation and reflective diaries, patterns of emotional communication emerged through gestures, rhythm, and shared sensory attention. A short video created from these

materials further articulated multisensory empathy as a communicative practice. Together, these observations provided a grounded understanding of non-verbal contexts in which human-plant interaction might meaningfully occur.

Semi-structured interviews, personas, and a stakeholder map were used to synthesize insights from individuals with autism, volunteers, and family members. This process clarified both shared and distinct needs across stakeholder groups and revealed the broader social and emotional ecosystem surrounding plant-integrated interaction. These findings guided the subsequent design focus towards a sensory communication system centered on shared feelings, co-creation, and playful interaction.

4.2 Co-creative Outcomes of Sensory and Plant-based Exploration

A co-creative workshop resulted in a collective fabric painting that combined participants’ emotions, sensory impressions, and interactions with plants. The collaborative making process established a shared communicative framework among participants, materials, and plants, reinforcing the role of co-creation in supporting non-verbal expression.

Patterns derived from the fabric paintings were digitized and later reinterpreted as core visual elements within the interactive system. These visual outcomes functioned as both aesthetic material and embodied traces of collective experience, forming a conceptual and visual foundation for the subsequent multisensory interaction design.

4.3 Multisensory Interactive System Design

Building on insights from user research and co-creation, the final installation translated plant-mediated capacitive interaction signals into a multisensory interactive system. Four indoor plants were incorporated, each connected to an MPR121 capacitive sensor and an Arduino board. Plants functioned as the primary interaction triggers, generating real-time visual and auditory responses based on human approach and touch (see Figure 3).

Each plant was assigned a distinct interactive configuration, combining visual patterns derived from the co-creative workshop, sounds recorded during field research, and short textual prompts. Subtle visual movement and ambient sound were triggered when participants approached a plant, while direct touch produced more dynamic visual rhythms, denser text displays, and stronger sound feedback. The textual elements were intentionally brief and poetic rather than instructional. They functioned as reflective cues rather



Figure 3: The final LeafLink installation and a close-up view of plant-mediated interaction, illustrating multisensory feedback triggered through human approach and touch.

than primary communication channels, supporting multisensory interpretation rather than replacing non-verbal interaction.

A key feature of the system was its support for simultaneous multi-point interaction. When multiple plants were engaged together, layered visual patterns, polyphonic sound textures, and combined textual responses emerged. These composite interactions reinforced the perception of plants as active participants within a shared communicative environment and foregrounded their role as more-than-human mediators in non-verbal interaction.

5 Evaluation and Preliminary Findings

A multidimensional evaluation was conducted to explore participants' experiences of the interactive system across eight self-reported dimensions, including emotional response, sensory engagement, comprehensibility, interaction smoothness, perceived communication with plants, reflection on human-nature relationships, playfulness, and educational inspiration. Across dimensions, mean scores ranged between 4.1 and 4.63, with comprehensibility and playfulness receiving the highest ratings.

Participants commonly described the interaction as relaxing, enjoyable, and immersive, noting that the system felt natural, fluent, and easy to understand. Many participants reported experiencing a sense of communication through touching and observing the plants, which prompted reflection on plant vitality and broader human-nature relationships. The installation was frequently characterized as playful and engaging, with participants expressing curiosity and a desire to continue exploring the interaction.

Rather than serving as a conclusive validation, these findings provide preliminary indications that plant-based interactive systems may support multisensory engagement, encourage emotional connection across species, and expand experiential possibilities within design-led HCI contexts.

6 Discussion

Observations from workshops highlighted the role of shared experience and collective engagement in plant-based interaction. Children often explored the installation together by touching multiple plants simultaneously, responding to emerging audiovisual feedback, and developing playful interaction sequences as a group. These interactions were characterized by joint attention and mutual curiosity, rather than task-oriented or verbally coordinated behavior.

Such observations suggest that plant-mediated interaction can support social engagement through co-presence and shared sensory reference points. By lowering communicative pressure and enabling side-by-side exploration, the system allowed participants to engage collectively without relying on explicit verbal communication. From a design perspective, this points to the potential of multispecies, non-verbal interaction to facilitate soft and open-ended forms of social connection in inclusive and therapeutic contexts.

7 Conclusion and Future Work

This work contributes (1) a clarified plant-mediated interaction framework, (2) a design-led methodology integrating co-creation and multisensory exploration in accessibility contexts, and (3) preliminary empirical insights into how plant-mediated interaction can support emotional expression through shared attention. By positioning plants as more-than-human mediators, the work demonstrates how calm, open-ended interaction can foster sensory engagement, shared attention, and gentle social connection. Rather than focusing on task efficiency or verbal communication, LeafLink contributes design insights into relational and experiential approaches to multispecies interaction within HCI.

Future work will further explore patterns of plant-mediated interaction in relation to human movement and touch dynamics to better understand how plant responses may relate to movement and emotional expression. Lightweight machine learning approaches may be explored to support adaptive interpretation of plant signals and more nuanced multimodal feedback. In addition, broader and more diverse user studies across different contexts will help examine how plant-based interaction is experienced by different groups, pointing towards more sensitive, personalised, and context-aware multispecies interactive systems.

References

- [1] Angelini, L., Caparrotta, S., Khaled, O.A. and Mugellini, E. 2016. EmotiPlant: Human-Plant Interaction for Older Adults. *Proceedings of the TEI '16: Tenth International Conference on Tangible, Embedded, and Embodied Interaction* (Eindhoven Netherlands, Feb. 2016), 373–379.
- [2] Barajas, A.O., Al Osman, H. and Shirmohammadi, S. 2017. A Serious Game for children with Autism Spectrum Disorder as a tool for play therapy. *2017 IEEE 5th International Conference on Serious Games and Applications for Health (SeGAH)* (Apr. 2017), 1–7.
- [3] Botros, F., Perin, C., Aseniero, B.A. and Carpendale, S. 2016. Go and Grow: Mapping Personal Data to a Living Plant. *Proceedings of the International Working Conference on Advanced Visual Interfaces* (Bari Italy, June 2016), 112–119.
- [4] Chien, J.T., Guimbretière, F.V., Rahman, T., Gay, G. and Matthews, M. 2015. Biogotchi!: An Exploration of Plant-Based Information Displays. *Proceedings of the 33rd Annual ACM Conference Extended Abstracts on Human Factors in Computing Systems* (Seoul Republic of Korea, Apr. 2015), 1139–1144.
- [5] Cho, J., Park, S., Jeon, B., Bae, B.-C. and Cho, J.-D. 2015. People's Emotional Responses to a Plant's Emotional Expression. *Proceedings of the Ninth International Conference on Tangible, Embedded, and Embodied Interaction* (New York, NY, USA, Jan. 2015), 545–550.
- [6] Evans, K. and Dubowski, J. 2001. Art Therapy with Children on the Autistic Spectrum: Beyond Words. (2001).
- [7] Fadi Hijab, M.H., Khattab, S., Al Aswadi, N., Bahameish, M., Qaraqe, M., Andres, M.J., Neves, J., Othman, A. and Al-Thani, D. 2024. Implementing a multi-sensory "Bag-of-stuff" in co-design sessions with autistic and non-autistic children. (July 2024), 60–71.
- [8] Febriantini, W.A., Fitriati, R. and Oktaviani, L. AN ANALYSIS OF VERBAL AND NON-VERBAL COMMUNICATION IN AUTISTIC CHILDREN. 2, 1.
- [9] Forlano, L. 2017. Posthumanism and Design. *She Ji: The Journal of Design, Economics, and Innovation*. 3, 1 (Mar. 2017), 16–29. <https://doi.org/10.1016/j.sheji.2017.08.001>.

- [10] Gabrys, J. 2016. Program Earth: Environmental Sensing Technology and the Making of a Computational Planet. U of Minnesota Press.
- [11] Giaccardi, E., Redström, J. and Nicenboim, I. 2025. The making(s) of more-than-human design: introduction to the special issue on more-than-human design and HCI. *Human-Computer Interaction*. 40, 1–4 (Jan. 2025), 1–16. <https://doi.org/10.1080/07370024.2024.2353357>.
- [12] Gil, A.F., Weinbeer, M. and Gloor, P.A. 2024. Can Plants Perceive Human Gestures? Using AI to Track Eurythmic Human-Plant Interaction. *Biomimetics*. 9, 5 (May 2024), 290. <https://doi.org/10.3390/biomimetics9050290>.
- [13] Hwang, S., Lee, K. and Yeo, W. 2010. My Green Pet: a current-based interactive plant for children. *Proceedings of the 9th International Conference on Interaction Design and Children* (New York, NY, USA, June 2010), 210–213.
- [14] Kumar, S. 2025. Dance Therapy as a Multisensory Stimulation Approach for Autism Spectrum Disorder (ASD). *Edumania-An International Multidisciplinary Journal*. 03, 01 (Jan. 2025), 145–154. <https://doi.org/10.59231/edumania/9102>.
- [15] Kuribayashi, S., Sakamoto, Y., Morihara, M. and Tanaka, H. 2007. Plantio: an interactive pot to augment plants' expressions. *Proceedings of the international conference on Advances in computer entertainment technology* (Salzburg Austria, June 2007), 139–142.
- [16] Mancini, C. 2011. Animal-computer interaction: a manifesto. *Interactions*. 18, 4 (July 2011), 69–73. <https://doi.org/10.1145/1978822.1978836>.
- [17] Mancini, C. 2013. Animal-computer interaction (ACI): changing perspective on HCI, participation and sustainability. *CHI '13 Extended Abstracts on Human Factors in Computing Systems* (Paris France, Apr. 2013), 2227–2236.
- [18] Marder, M., Vattimo, G. and Zabala, S. 2013. *Plant-Thinking: A Philosophy of Vegetal Life*. Columbia University Press.
- [19] Pouya, S., Bayındır, E. and Pouya, S. 2024. Sensory garden design proposal for children with autism spectrum disorder. *Support for Learning*. 39, 1 (2024), 44–57. <https://doi.org/10.1111/1467-9604.12466>.
- [20] Sareen, H., Zheng, J. and Maes, P. 2019. Cyborg Botany: Augmented Plants as Sensors, Displays and Actuators. *Extended Abstracts of the 2019 CHI Conference on Human Factors in Computing Systems* (Glasgow Scotland Uk, May 2019), 1–2.
- [21] Seo, J.H., Sungkajun, A. and Suh, J. 2015. Touchology: Towards Interactive Plant Design for Children with Autism and Older Adults in Senior Housing. *Proceedings of the 33rd Annual ACM Conference Extended Abstracts on Human Factors in Computing Systems* (Seoul Republic of Korea, Apr. 2015), 893–898.
- [22] Steer, C., Robinson, S. and Jones, M. 2015. Growth, Change and Decay: Plants and Interaction Possibilities. *Proceedings of the 33rd Annual ACM Conference Extended Abstracts on Human Factors in Computing Systems* (Seoul Republic of Korea, Apr. 2015), 2037–2042.
- [23] Tao, S. and Lioret, A. 2016. Plant feedback on environmental changes and human interactions. *SIGGRAPH ASIA 2016 Posters* (Macau, Nov. 2016), 1–2.
- [24] Westerlaken, M. and Gualeni, S. 2016. Becoming with: towards the inclusion of animals as participants in design processes. *Proceedings of the Third International Conference on Animal-Computer Interaction* (Milton Keynes United Kingdom, Nov. 2016), 1–10.
- [25] 2016. Guidelines for the undergraduate psychology major: Version 2.0. *American Psychologist*. 71, 2 (2016), 102–111. <https://doi.org/10.1037/a0037562>.